

# RAJAT SAVDEKAR

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## EDUCATION

**University of Southern California, Los Angeles**  
Master of Science, Computer Science

Aug 2024 – May 2026  
GPA: 3.63

**University of Mumbai, Mumbai**  
Bachelor of Engineering, Computer Engineering

Aug 2019 – May 2023  
GPA: 3.7

## SKILLS

**Languages:** Python, TypeScript, JavaScript, Java, C++, SQL

**Backend & APIs:** Node.js, Flask, FastAPI, REST APIs, WebSockets

**AI:** LLMs, RAG, OpenAI Whisper, prompt engineering

**Data & Messaging:** PostgreSQL, MySQL, MongoDB, Redis, event sourcing

**Cloud & Infrastructure:** AWS, Docker, Docker Compose, CI/CD, Git, Next.js, React

## EXPERIENCE

### Software Developer – Systems & Performance

Aug 2026 – Present

Easley-Dunn Productions, Inc.

Torrance, CA

- Reduced AI-narration inference cost by **60%** across **150+ surgical replays** by 4-bit quantizing a 7B medical LLM, batching speech synthesis, and caching model plus translation outputs.
- Locked Quest training scenes at **72 Hz** and cut drill frame time **4x** by stripping 255 extra mesh-deform scripts and jobifying vertex updates, then added CI/CD that deploys Quest builds, caught **8** regressions before merge.
- Cut replay upload failures by replacing **100 MB** in-memory SQLite blobs with streamed object storage and a file API that matches the Quest client.

### Software Engineer

May 2023 – May 2024

Jalgaon Fruit Sales Cooperative Ltd – Agritech Supply Chain Platform

Jalgaon, India

- Decomposed a legacy PHP, SQL monolith by building Python Flask microservices backed by PostgreSQL, orchestrating automated cron jobs to synchronize state and cutting query latency by **40%** across **1,000+ daily truck transactions**.
- Doubled peak harvest write throughput (**2x**) and scaled revenue from **\$100K to \$200K** by architecting an isolated write path that eliminated 10–15 minute freezes from MyISAM table locking.
- Reduced support chatbot inference costs by **35%** across inventory lookups by implementing Redis-based semantic prompt caching, model fallback routing, and token compression over warehouse metadata.
- Delivered a web procurement platform with RESTful APIs and live analytics dashboards for **150+ daily clerks** across 5 warehouses, reducing deploy cycles by **30%** by containerizing services with Docker, pytest validation, and CI/CD.

## SELECTED PROJECTS & OPEN SOURCE

### MockPad – Real-Time Collaborative Engine & Multimodal AI Replay

[\[Live\]](#) [\[GitHub\]](#)

- Architected a real-time collaborative editor on Yjs CRDTs over custom WebSockets, sustaining **sub-100ms sync latency** for **30+ active users** deployed on a single \$6/month DigitalOcean droplet.
- Cut server peak memory consumption by **80%** with **zero session data loss** by implementing an LRU disk-eviction strategy to LevelDB for inactive collaborative rooms.

### CloudBridge – AI Cloud Architecture Advisor & Topology Engine

[\[GitHub\]](#)

- Engineered an LLM-driven synthesis engine in FastAPI mapping natural language constraints into validated AWS topologies with IAM guardrails and automated Terraform export against the **Well-Architected Framework**.
- Built a reactive DAG topology generator in React Flow and TypeScript, delivering **sub-100ms** real-time parameter mutations, cost projections, and multi-service dependency recalculations.

### DevStack – Open Source Developer Infrastructure Platform

[\[GitHub\]](#)

- Engineered a durable workflow execution engine on an event-sourced PostgreSQL schema with TypeScript deterministic replay, restoring crash state **without re-triggering side effects**.
- Built a distributed feature-flag targeting engine delivering **sub-5ms evaluation latency** via Redis caching and real-time WebSockets, secured with RS256 JWT auth and a federated GraphQL gateway.

## HONORS & PUBLICATIONS

**Aura – visionOS Accessibility Assistant.** Aug 2025. 2nd place of 24-hour hackathon; built with Swift, SwiftUI, and RealityKit, delivering fully on-device live captions and environmental sound awareness. [\[GitHub\]](#)

**Research Publications.** *Texture Feature Analysis Using GLCM Matrix (10 citations)*; medical-transcript NLP; LSB image steganography. [\[GitHub\]](#)